Can Claude Count Letters? — Lab Report

TL;DR: Answering instantly, Claude miscounts letters, worst of all claude-haiku-4-5 on Hebrew (55% correct); a step-by-step (CoT) prompt fixed almost every mistake (99.9% over 1,600 runs).

Experiment. Each run takes a random word (from the 20k most common Hebrew or English subtitle words) and a random letter from it, and asks a fresh Claude Code session: How many times does the letter "r" appear in the word "reports"? Settings: claude -p --tools "" --strict-mcp-config --system-prompt "You are a helpful assistant.". 5 factors × 2 values = 32 setups × 100 runs (the same 100 words in each): model claude-haiku-4-5 vs claude-haiku-5-5; Hebrew vs English words; question in Hebrew vs English; no CoT ("Answer with a number only.") vs CoT ("Think step by step: spell the word out one letter at a time and count. On the last line write "Answer: <number>"."); thinking off (MAX_THINKING_TOKENS=0) vs on (default). Correct = final number equals the true count (for Hebrew final letters, ם counted as מ or not are both accepted).

My expectations vs. results (no CoT, thinking off unless noted; runs per side in brackets):

Expectation	Result	Verdict
CoT beats no CoT	99.9% vs 85.4% (800), thinking off	Confirmed
English question beats Hebrew question	87.8% vs 83.0% (400)	Partly: Hebrew words only (88.5% vs 77.5%, 200); haiku-5-5 was better in Hebrew
Hebrew Q on English words < English Q on Hebrew words	88.5% vs 88.5% (200)	No difference
Newer model beats older	haiku-5-5 95.8% vs haiku-4-5 75.0% (400)	Confirmed

Harder cases. Thinking fixed claude-haiku-4-5 (100%, 400 runs) but not claude-haiku-5-5, which repeated the same mistakes on English words (89%, 100 runs), suggesting it doesn't think by default on easy questions. Almost all mistakes were off by one.